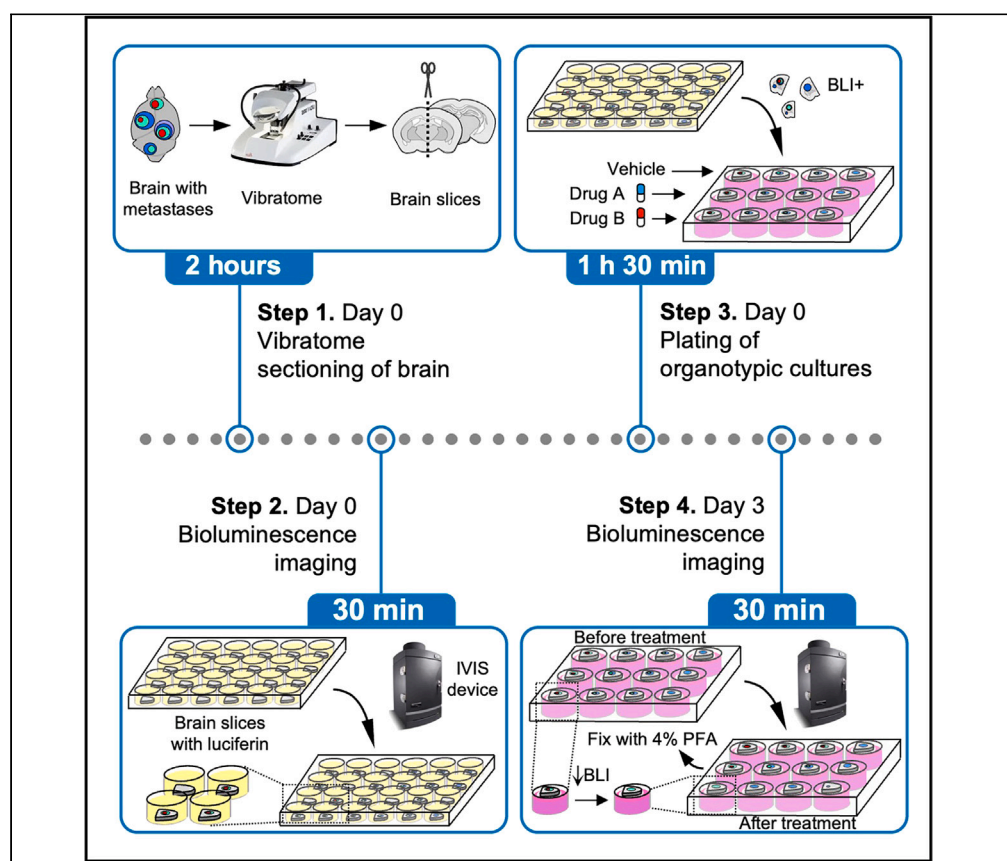


Protocol

Protocol to generate murine organotypic brain cultures for drug screening and evaluation of anti-metastatic efficacy



Organotypic brain cultures are short-term assays that phenotypically and functionally recapitulate brain metastatic cells *in vivo*. Here, we present a protocol to generate murine organotypic brain cultures for drug screening. We describe steps for sectioning of murine brains and plating of organotypic cultures. We then detail evaluation of the anti-metastatic effect of chemical compounds through bioluminescence imaging before and after drug treatment. Combined with downstream applications, this protocol allows comprehensive characterizations of both cancer cells and the tumor-associated microenvironment.

Publisher's note: Undertaking any experimental protocol requires adherence to local institutional guidelines for laboratory safety and ethics.

Lucía Zhu, Lauritz Miarka, Patricia Baena, María Perea-García, Manuel Valiente

mvaliente@cnio.es

Highlights

Protocol to generate organotypic brain cultures for drug screening

Steps for mouse brain sectioning and plating of organotypic cultures

Bioluminescence imaging before and after drug treatment to assess metastatic cells

Fast and simple readouts for scoring anti-metastatic efficacy of drugs tested

Zhu et al., STAR Protocols 4, 102194
June 16, 2023 © 2023 The Author(s).
<https://doi.org/10.1016/j.xpro.2023.102194>



Protocol

Protocol to generate murine organotypic brain cultures for drug screening and evaluation of anti-metastatic efficacy

Lucía Zhu,¹ Lauritz Miarka,¹ Patricia Baena,¹ María Perea-García,¹ and Manuel Valiente^{1,2,3,*}¹Brain Metastasis Group, Spanish National Cancer Research Center (CNIO), 28039 Madrid, Spain²Technical contact³Lead contact*Correspondence: mvaliente@cnio.es<https://doi.org/10.1016/j.xpro.2023.102194>

SUMMARY

Organotypic brain cultures are short-term assays that phenotypically and functionally recapitulate brain metastatic cells *in vivo*. Here, we present a protocol to generate murine organotypic brain cultures for drug screening. We describe steps for sectioning of murine brains and plating of organotypic cultures. We then detail evaluation of the anti-metastatic effect of chemical compounds through bioluminescence imaging before and after drug treatment. Combined with downstream applications, this protocol allows comprehensive characterizations of both cancer cells and the tumor-associated microenvironment. For complete details on the use and execution of this protocol, please refer to Zhu et al. (2022).¹

BEFORE YOU BEGIN

The protocol below describes the specific steps to obtain organotypic cultures of established brain metastases from murine brains, which can be used to probe the anti-metastatic potential of chemical compounds. Here, we use the brain-metastatic (BrM) derivative of the human lung adenocarcinoma cell line H2030 (H2030-BrM). However, we have successfully used this protocol with other BrM cell lines of both human and mouse origin, for example from breast cancer (E0771-BrM) or melanoma (B16/F10-BrM).

Note: The cells used in this protocol are engineered to express firefly luciferase and the green fluorescent protein (GFP) as described elsewhere,² which allows non-invasive tracking of metastasis growth *in vivo* and *ex vivo* as well as unambiguous identification of tumor cells in histological sections, respectively.

Institutional permissions

This protocol requires the use of mouse tissue. All animal experiments should be performed in accordance with relevant governmental and institutional guidelines. All experiments described here were approved by the CNIO (IACUC.030-2015), Instituto de Salud Carlos III (CBA35_2015-v2), and Comunidad de Madrid Institutional Animal Care and Use Committee (PROEX250/15 and PROEX135/19).

Generation of established brain metastasis *in vivo*

⌚ Timing: 5–7 weeks

⌚ Timing: 60–90 min (for step 1)



⌚ Timing: 15 min per mouse (for step 2)

⌚ Timing: 2–5 min per 3–5 mice, depending on capacity of the IVIS Xenogen imaging device (for step 3)

Note: While planning an experiment using this protocol, it is important to consider that one mouse brain with established metastasis will yield approximately 20–40 organotypic cultures. We recommend using 4–6 organotypic cultures per experimental condition, therefore we recommend planning the number of mice needed accordingly. If correctly performed, 100% of mice inoculated with H2030-BrM cells develop established brain metastasis after 5–7 weeks.

1. Preparation of H2030-BrM cells for intracardiac injection.
 - a. Prepare the medium used for culturing H2030-BrM cells (see [materials and equipment](#)).
 - b. Culture H2030-BrM cells in a 10 or 15 cm plate under adherent conditions using supplemented RPMI medium.
 - c. On the day of the planned injection into mice, cells should have reached a confluency of 70%–80% in the culturing plate. Wash plate with sterile PBS.
 - d. Detach cells by adding the appropriate volume of trypsin-EDTA solution (0.05%) (1 mL for a 10 cm plate or 2 mL for a 15 cm plate).
 - e. Incubate them at 37° Celsius for 5–10 min.
 - f. To stop trypsinization, add supplemented RPMI medium to the plate (3 mL for a 10 cm plate or 6 mL for a 15 cm plate).
 - g. Collect detached cells in 15 mL conical tubes and centrifuge cells at $1,500 \times g$ for 5 min.
 - h. Discard supernatant and add fresh medium.
 - i. Count viable cells using the trypan blue exclusion method and a Neubauer counting chamber (or similar).
 - j. Take the amount of cell suspension that contain 10^5 cells per mouse planned for injection and centrifuge at $1,500 \times g$ for 5 min.

Note: If more than one mouse is planned, centrifuge all the cells together. It is convenient to prepare at least 10 times the number of cells needed per mouse since the process of loading the syringe with cells and the intracardiac inoculation could be inefficient (i.e., failure to inject in heart). Consequently, having access to an extra amount of cell suspension might help to avoid running out of cells derived from the situations described above. The approximate number of H2030-BrM cells collected from a 70%–80% confluent 10 cm and 15 cm plate are 3 and 8×10^6 , respectively. We recommend culturing as many plates as needed depending on the number of mice planned for injection and after accounting for the extra number of cells suggested above.

- k. Discard the supernatant and resuspend your cell pellet in sterile PBS to a concentration of 10^5 in 100 μ L.

Note: Following the recommendation above about preparing at least 10 times the number of cells needed, a total of 10^6 cells (resuspended in 1 mL of PBS) per mouse planned for injection should be available after resuspension.

- l. Prepare multiple Eppendorf tubes with 1 mL of your cell suspension.
 - m. Place tubes on ice.
2. Intracardiac inoculation of BrM cells into athymic nu/nu mice.

Note: The mouse strain required is dependent on the species of origin of the cell line used. For the model described here, we use athymic nu/nu mice 4–10 weeks of age. For other cell lines

as described above, different mouse strains, such as C57BL/6 mice, might be required. All mice are maintained under housing and husbandry conditions accredited by the Association for Assessment and Accreditation of Laboratory Animal Care (AAALAC) International. Mice are specific pathogen-free (SPF) with microbiological and environmental parameters constantly monitored.

Note: In this protocol we use an induction chamber and a heating pad with nose cones to anesthetize mice with 3% isoflurane. If this equipment is not available, mice can also be anesthetized using other methods, such as intraperitoneal injection of ketamine (100 mg/kg)/xylazine (10 mg/kg). However, we strongly recommend the first method of anesthesia as it is easier to modulate and control, thus maximizing the efficacy of the intracardiac procedure.

- a. Clean biosafety cabinet, heating pad and anesthesia induction chamber with 70% ethanol.
- b. Following aseptic procedures, introduce the cooled cells, heating pad, syringe with needles and small pieces of paper tape into the biosafety cabinet.

Note: The cells should be kept cool to maintain viability until the moment of use, either by storing them in a fridge close to the biosafety cabinet or by keeping them on ice. Under no circumstances the solution can freeze down as this will damage the cancer cells.

- c. Set up a housing cage under a heating lamp at appropriate distance, which will be used for mice to recover after anesthesia.
- d. Set temperature of the heating pad to 37° Celsius.
- e. Anesthetize one mouse at a time by placing them in the induction chamber using 3% isoflurane.
- f. When mice are appropriately anesthetized, as assessed by loss of toe pinch response, place one mouse on the heating pad with its nose in the nose cone.

Note: The animal should be on its back facing up with its fore limbs fixated carefully using paper tape pieces.

- g. Disinfect the chest of the mouse using 70% of ethanol.
- h. Using a syringe with a 26G needle, mix the cell suspension in the Eppendorf tube.
- i. Draw 500 µL of the cell suspension.

△ CRITICAL: Make sure no bubbles are present. If bubbles are present after loading the syringe repeatedly hit the syringe with a finger until the bubbles are gone always maintaining the needle towards the floor.

- j. Vertically insert the needle into the chest of the animal to reach the heart.

Note: As a reference you should insert the needle into the left side of the thoracic cavity close to the midline in the 4th intercostal space.

Note: Ensure that a strong, pulse-synchronous backflow of blood into the syringe is present. A weaker backflow indicates the positioning of the needle in the right ventricle. Injecting the cell suspension into the right ventricle will lead to a strong dissemination to the lungs, rather than systemically, therefore we recommend discarding the mouse.

- k. Slowly inject 100 µL and equally carefully remove needle in a straight position.²
- l. Remove mouse from the heating pad and place it under a heating lamp in order to recover from anesthesia.

Note: Manipulate the mouse slowly after the injection.

- m. House mice under pre-injection conditions after recovery from anesthesia. No analgesia nor post-operative care are required.
- 3. Bioluminescence imaging (BLI) to track development of metastatic disease non-invasively.

Note: BLI should be performed right after finishing intracardiac injection of all mice to confirm correct inoculation. Afterwards weekly measurements should be carried out.

- a. Thaw an appropriate amount (100 μ L / mouse) of D-luciferin (15 mg/mL) at room temperature (19°–21° Celsius).
- b. According to the capacity of the imaging device, anesthetize a number of mice by placing them in the induction chamber using 3% isoflurane.
- c. When mice are appropriately anesthetized (as assessed by loss of toe pinch response), inject 100 μ L D-luciferin into the retro-orbital sinus using a syringe with a 30G needle.³
- d. Place mice noses into the nose cones of the imaging device. The mice should be on their stomach, facing down.
- e. Depending on the progression of metastatic disease, acquire images starting at an exposure time of 120 s.

Note: If images are saturated, reduce exposure time in intervals of 60, 30, 10 and 1 s until the image is not saturated.

- f. Using the image analysis software Living Image 4.5, select the head of each animal as a region of interest (ROI) and measure the photons/seconds/cm²/steradian.

△ **CRITICAL:** Mice are considered ready for the organotypic assay once the value of the head reaches 5×10^6 and higher and can be processed at any time starting from that moment, until they reach endpoint (generally 3–7 days after they reach the value).

Note: During an experiment, keep exactly the same ROI size to measure BLI signal. ROIs can be saved in the first image measured and re-loaded in subsequent images.

Note: The model described in this protocol usually reaches the endpoint at 5–7 weeks post-injection. However, tumor growth dynamics can be heterogenous, so weekly BLI measurements are critical to ensure the right timing of each animal, although it is frequent that more than one animal reach the value and can be processed simultaneously. In addition, different models might not reach this value (i.e., syngeneic models) so this reference must be adapted to the experimental models used.

- g. After acquisition, remove animals from the imaging device and let them recover from the anesthesia.
- h. House mice under pre-imaging conditions.

Preparing the low-melting agarose

⌚ **Timing:** 30 min

- 4. On the day of the experiment and prior to processing the brains, heat a water bath to 42° Celsius.
- 5. Dissolve 1 g of low-melting agarose powder in 25 mL (4%) of ice-cold HBSSsup.

Note: The amount of 4% low-melting agarose to prepare depends on the number of brains intended to be cut. Here, 25 mL will suffice for 2–3 brains. We recommend preparing 4% low-melting agarose in excess to account for errors during the embedding.

6. Gently shake the beaker containing both ingredients.
7. Facilitate solvation by heating the beaker in the microwave until boiling three consecutive times, to ensure agarose is well solved.
8. Leave the 4% low-melting agarose in the pre-heated water bath for at least 30 min in order for it to reach a stable temperature.

Processing of mouse brains

⌚ Timing: 30 min per mouse

9. Following the same procedure as for the *in vivo* BLI described above, anesthetize one mouse at a time.
10. Inject 100 μ L of D-luciferin retro-orbitally.

⚠ **CRITICAL:** It is critical to inject D-luciferin retro-orbitally³ while the mouse is alive to be able to perform *ex vivo* BLI measurement.

Note: If *in vivo* BLI was performed right before, new D-luciferin should still be injected to ensure sufficient amount in the animal for the *ex vivo* BLI measurement.

11. Euthanize the mouse using cervical dislocation.
12. Harvest the brain from the mouse.
 - a. Use a pair of hardened fine scissors to carefully cut upward along the sagittal suture starting from the brain stem.
 - b. Using a Dumont forceps, peel away the two halves of the skull to expose the brain.
 - c. Carefully isolate the brain.
13. Place the brain on a 10 cm plate.
14. Perform *ex vivo* BLI measurement in order to locate metastatic lesions.
15. Acquire images starting at an exposure time of 120 s.

Note: If images are saturated, reduce exposure time in intervals of 60, 30, 10 and 1 s until the image is not saturated.

Optional: Using Living Image 4.5, select the brain as ROI and measure the photons/seconds/ cm^2 /steradian. A brain with a value of 5×10^6 photons/seconds/ cm^2 /steradian or higher will yield approximately 20–40 organotypic cultures. Generally, brains that reach this value have multiple metastases spread throughout and allow to obtain a high number of BLI positive brain slices after vibratome sectioning. Whereas brains not reaching this value or that reach the value but present with a large single tumor, will yield a smaller number of organotypic cultures due to less tumor foci. However, these brains are still eligible for the experiment and should not be discarded.

16. Post BLI, immediately transfer brains to ice-cold HBSSsup. They are now ready to be embedded.

KEY RESOURCES TABLE

REAGENT or RESOURCE	SOURCE	IDENTIFIER
Chemicals, peptides, and recombinant proteins		
RPMI-1640 medium	Sigma-Aldrich	Cat# R8758
Penicillin/Streptomycin	Life Technologies	Cat# 15-140-122
DMEM medium	Lonza	N/A

(Continued on next page)

Continued		
REAGENT or RESOURCE	SOURCE	IDENTIFIER
Trypsin-EDTA (0.05%)	Gibco	Cat# 25300096
20% Paraformaldehyde	Aname	Cat# 15713
L-Glutamine	Gibco	Cat# 25030-024
Amphotericin B	Fisher Scientific	Cat# 15290018
Fetal bovine serum	Life Technologies	Cat# 10500064
D-Luciferin	Syd Labs	Cat# MB000102-R70170
HBSS 10x	Gibco	Cat# 4185052
Nusieve GTG Agarose	Lonza	Cat# 50080
Hepes	Sigma-Aldrich	Cat# H3375-100G
D-Glucose	Sigma-Aldrich	Cat# G7528
Isoflurane	Isoflo®	N/A
Dulbecco's Phosphate Buffered Saline (PBS)	Sigma-Aldrich	Cat# D8537-500ML
Experimental models: Cell lines		
H2030-BrM human lung adenocarcinoma cell line (1–20 passages after thawing)	Nguyen et al. ⁴	N/A
Experimental models: Organisms/strains		
Female mice: Hsd:Athymic Nude-Foxn1 ^{nu} (5 weeks)	Envigo	Cat# 069
Software and algorithms		
Living Image software, v4.5	Perkin-Elmer	https://www.perkinelmer.com/de/lab-products-and-services/resources/in-vivo-imaging-software-downloads.html#LivingImage
Other		
Needle 26Gx 1/2	Fisher Scientific	Cat# AN*2613R1
Needle 30G	Fisher Scientific	Cat# AN*3013R1
12-well Clear TC-treated plates	Corning	Cat# 3513
24-well Clear TC-treated plates	Corning	Cat# 3526
Loctite 401 Instant adhesive glue	Henkel Adhesives	123011
Peel away disposable embedding molds	Electron Microscopy Sciences	Cat# 70182
Double edge stainless razor blades	Electron Microscopy Sciences	Cat# 72000
Flat spatula	Fine Science Tools	Cat# 10090-17
Dumont forceps	Fine Science Tools	Cat# 11295-20
Hardened fine scissors	Fine Science Tools	Cat# 14090-09
0.8 µm Whatman Nuclepore Track-Etched Membranes	Sigma-Aldrich	Cat# WHA110409
Sterile vacuum filter/bottle system (0.22 µm pore)	Corning	Cat# 430758
Leica VT1000 S Vibratome	Leica	Cat# 1404723512
IVIS Xenogen machine	Caliper Life Sciences	N/A
Heating pad	Cibertec	N/A
Induction chamber	Plexx	Sure-Seal Mouse-Rat Chamber
N/A: not available/applicable.		

MATERIALS AND EQUIPMENT

Supplemented RPMI medium (for cell culture of H2030-BrM)		
Reagent	Final concentration	Amount
RPMI-1640	87.5% (v/v)	438 mL
Fetal bovine serum	10% (v/v)	50 mL
L-Glutamine	2 mM	5 mL

(Continued on next page)

Continued

Reagent	Final concentration	Amount
Penicillin/Streptomycin	100 IU/mL	5 mL
Amphotericin B	1 µg/mL	2 mL
Total		500 mL

Prepare in a sterile environment, store at 4° Celsius, do not use longer than 3 months.

Supplemented Hank's buffered salt solution (HBSSsup)

Reagent	Final concentration	Amount
HBSS 10×	1×	50 mL
1 M Hepes (pH 7.4)	2.5 mM	1.25 mL
1 M D-Glucose	30 mM	15 mL
100 mM CaCl ₂ · 2H ₂ O	1 mM	5 mL
100 mM MgCl ₂ · 6H ₂ O	1 mM	5 mL
1 M NaHCO ₃	4 mM	2 mL
H ₂ O	84.35% (v/v)	421.75 mL
Total		500 mL

Prepare in a sterile environment, store at 4° Celsius, do not use longer than 7 days.

Slice culture medium (SCM)

Reagent	Final concentration	Amount
DMEM medium	70% (v/v)	210 mL
HBSSsup	25.8% (v/v)	77.4 mL
1 M D-Glucose	27 mM	8.1 mL
L-Glutamine	1 mM	1.5 mL
Penicillin/Streptomycin	100 IU/mL	3 mL
Fetal bovine serum	5% (v/v)	15 mL
Total		300 mL

Prepare in a sterile environment, store at 4° Celsius, do not use longer than 1 month.

Reagents needed for preparation of relevant media:

- D-Glucose (1 M): 18.016 g in 100 mL H₂O.
- CaCl₂ · 2H₂O (100 mM): 1.47 g in 100 mL H₂O.
- MgCl₂ · 6H₂O (100 mM): 2.03 g in 100 mL H₂O.
- NaHCO₃ (1 M): 8.4 g in 100 mL H₂O.

Prepare in a sterile environment, store at 4° Celsius, do not use longer than the expiry date of the reagents.

D-Luciferin potassium salt preparation

Working in a biosafety cabinet without light, dissolve 1 g of D-Luciferin in 66.6 mL of sterile PBS to get a final concentration of 15 mg/mL. Once solved, filter the solution using a 0.22 µm filter.

Aliquot the solution and store at −20° Celsius covered from light, do not use longer than the expiry date.

STEP-BY-STEP METHOD DETAILS

Vibratome sectioning of brain

⌚ Timing: 2 h

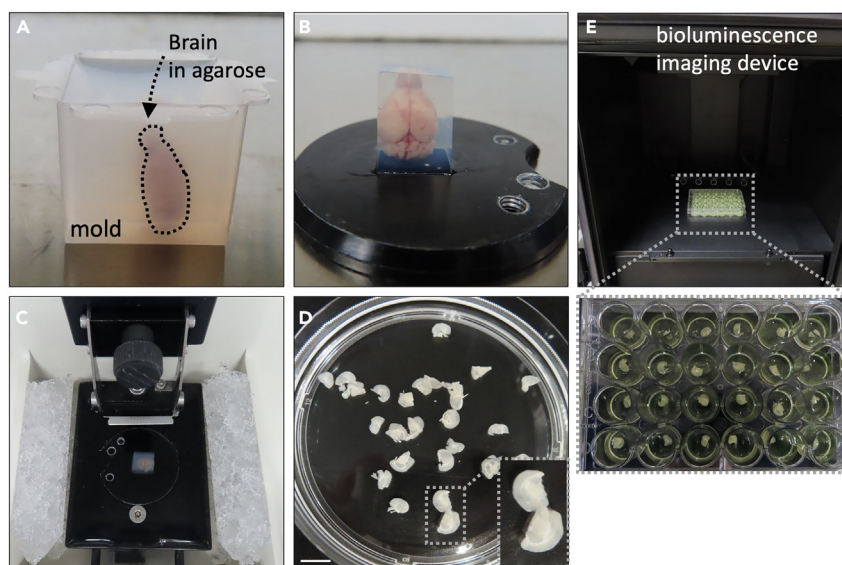


Figure 1. Obtaining brain slices in the vibratome

- (A) Embedded brain in low-melting agarose and peel away molds.
 (B) Trimmed agarose block is fixed to vibratome platform in coronal position using glue.
 (C) The platform including the brain is fixed inside the vibratome buffer tray and covered entirely by ice-cold HBSSsup.
 (D) Divided brains stored in ice-cold HBSSsup. Scale bar: 10 mm.
 (E) Brain slices in single wells of a 24-well plate for BLI.

This section describes how the mouse brain with metastases is embedded in an agarose block and sectioned into slices using a vibratome. All steps are performed under a laminar flow cabinet.

1. Embed brain in 4% low melting agarose preheated at 42° Celsius ([Methods video S1](#) and [Figure 1A](#)).
 - a. Transfer brain to a 10 cm plate with HBSSsup.
 - b. Dry the brain by holding it with a forceps and gently tapping it in a dry 10 cm plate.
 - c. Fill the Peel Away Disposable Embedding Mold with 4% low melting agarose preheated at 42° Celsius.
 - d. Immerse brain in the mold with 4% agarose.
 - e. Ensure full coverage of the brain surface with agarose by gently stirring the melted agarose with a 20–200 μ L tip around the brain ([Methods video S1](#)).
 - f. Place the mold on ice and use the tip to gently position the brain in coronal position with the region of most interest upwards. [Troubleshooting 1](#).

Note: The region of interest of each brain depends on the location of the metastases and can be selected based on the corresponding ex vivo BLI image.

- g. Let the agarose solidify.
2. Section the edges and the walls of the mold with a razor blade to extract the agarose block from the Peel Away Disposable Embedding Mold ([Methods video S2](#)).

△ CRITICAL: In steps 2 and 3, be careful to not touch the brain with the razor blade to avoid damage of the tissue.

3. Trim the excess of agarose with the razor blade ([Methods video S2](#)).

△ **CRITICAL:** When performing step 3, ensure that the agarose is covering the brain in all sides of the block, and no brain surface is exposed.

4. Fixate the trimmed agarose block with the brain inside to the sample holder using instant adhesive glue. The brain is placed in coronal position with the region of most interest upwards ([Figure 1B](#)).

Note: If properly trimmed, up to 4–5 brains can be simultaneously placed on the sample holder and sectioned.

5. Place the sample holder inside the buffer tray and fill it with ice-cold HBSSsup ([Figure 1C](#)).
 - a. Keep the buffer tray cold by surrounding it with ice.
6. Place half of a double-edge razor in the razor-holding arm of the vibratome.
7. Use the control panel to fix the settings and initiate sectioning.
 - a. Set the speed (2 mm/s), frequency (5 Hz), and thickness of the slices (250 μm).
 - b. Move the sectioning arm using the forward-reverse option in the control panel to set the initial and endpoint of each section ([Methods video S3](#)).
 - c. Select the continuous mode (*Cont*) and press *Start*.
 - d. The vibratome will automatically section the brain based on the settings. [Troubleshooting 2](#).

Note: In step 7b, maximum speed (*V-max*) can be selected to set the initial and endpoints. However, position the buffer tray using the up-down option in the control panel to maintain the razor above the level of the agarose and ensure not sectioning the brain during this process. Remember to unselect *V-max* before pressing *Start*.

8. Collect brain slices with a spatula and divide each coronal slice into two halves with a needle tip ([Methods video S3](#)).
 - a. Keep brain slices in a 10 cm plate with ice-cold HBSSsup and on ice ([Figure 1D](#)).

Note: Collect as many brain slices as needed to ensure that all regions of the brain with metastases, based on the *ex vivo* BLI image, have been cut. One whole brain yields approximately 25–30 coronal brain slices (before cutting into halves).

Bioluminescence imaging (day 0) of brain slices

⌚ **Timing:** 30 min

In this section, brain slices with positive bioluminescence signal are selected.

9. Transfer the halves of the coronal brain slices (referred to as brain slices hereafter) to a 24-well plate with HBSSsup.
 - a. Pipette 1 mL of HBSSsup per well.
 - b. Place one brain slice per well.

Note: Step 9 is performed under a laminar flow cabinet. Keep plates on ice while working.

- c. Add 20 μL of D-luciferin (15 mg/mL) per mL of HBSSsup, gently shake the plate to mix, and incubate 5 min at room temperature (19°–21° Celsius) in the dark.

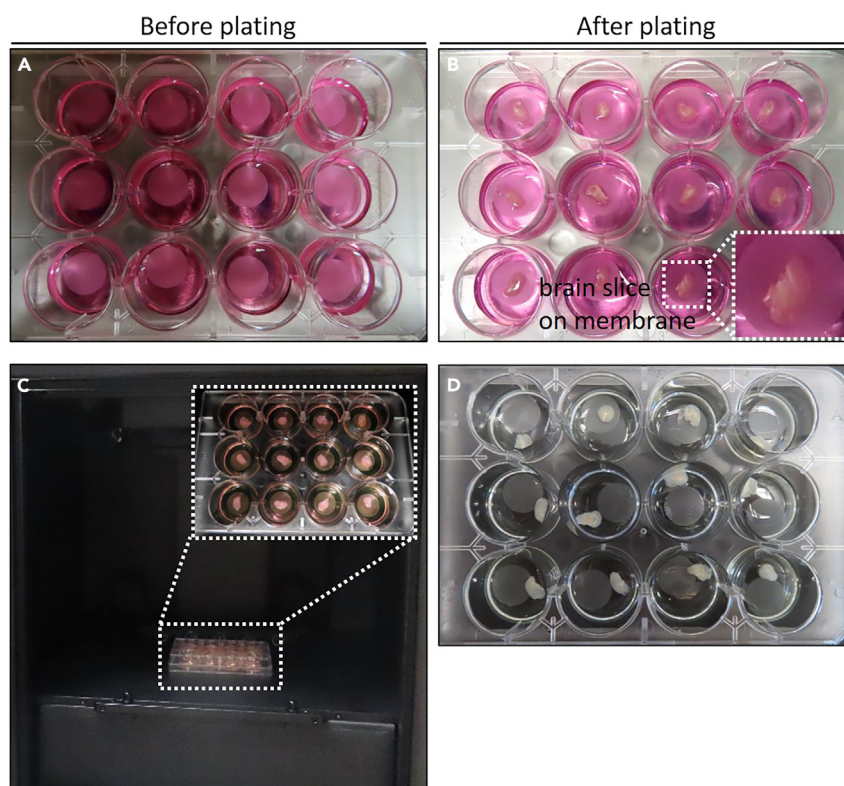


Figure 2. Plating, imaging, and fixing brain slices

- (A) Membranes floating on SCM.
(B) Brain slices placed wrinkle-free on membranes.
(C) Plate with brain slices and luciferin in BLI device for measurement.
(D) Fixed brain slices, using 4% PFA.

10. Acquire BLI images at exposure times of 10 and 1 s. If less than half of the total number of slices are positive, repeat the acquisition at higher exposure times (30, 60, or 120 s) (Figure 1E).

Note: Keep plates with brain slices on ice after BLI acquisition until step 14.

11. Measure the BLI signal of each brain slice using the image analysis software Living Image 4.5, and categorize BLI positive brain slices as with high, medium, or low BLI signal.

Note: Categorization of BLI positive brain slices by level of signal will allow to equally distribute high, medium, and low BLI signal brain slices across different treatment conditions when plating and avoid bias.

Brain slices plating and culture

⌚ **Timing:** 1 h 30 min

In this section, BLI positive brain slices are placed on top of membranes and cultured, resulting in organotypic cultures. All steps are performed under a biosafety cabinet in an equipped room for primary cell culture.

12. Prepare 12-well plates with slice culture medium (SCM) and chemical compounds of interest (Figure 2A) (examples of chemical compounds used are extensively provided in our previous publication by Zhu et al.¹).

- a. Per treatment condition, prepare a mix of SCM with the appropriate volume of chemical compound to achieve the desired final concentration.
 - i. Prepare the mix for the corresponding control (i.e., SCM + vehicle) condition.
 - b. Add 1 mL of mix per well in a 12-well plate.
 - c. Using a Dumont forceps, place one 0.8 μ m pore membrane per well on top of the liquid leaving the matte side of the membrane upwards.
13. Transfer BLI positive brain slices with a spatula from the 24-well plates to a 10 cm plate with ice-cold HBSSsup to facilitate manipulation.

△ CRITICAL: Transfer one slice at a time and annotate the well of origin (in the 24-well plate) and the well where it is plated (in the 12-well plate). This is critical for the subsequent BLI analysis.

14. Collect the brain slice with a spatula and gently place it on top of a membrane in the 12-well plate (1 slice per membrane) assisted by another spatula ([Methods video S4](#) and [Figure 2B](#)).
 - a. Repeat until the plate is completed.

△ CRITICAL: Avoid wrinkles in the slice as they will remain in the organotypic culture for the subsequent applications (e.g., histological analysis). [Troubleshooting 3](#).

15. Gently transfer the plate to a CO₂ incubator and ensure all membranes (with their respective slices) are floating. [Troubleshooting 4](#).
16. Incubate at 37° Celsius and 5% CO₂ for 72 h (3 days).

⏸ Pause point: 72 h.

Optional: Organotypic cultures can be maintained up to 7 days in culture in a CO₂ incubator. In this case, SCM with respective chemical compounds should be refreshed at day 3, by removing old medium with a pipette and gently adding new medium on the walls of the wells. Do not use a vacuum pump to remove old medium and ensure to not wet the upper surface of the membranes when adding new medium. If correctly done, membranes with their respective organotypic cultures on top should float with the addition of new medium.

Bioluminescence imaging (day 3) and fixation of organotypic cultures

⌚ Timing: 30 min

In this section, cultured brain slices or organotypic cultures are subjected to BLI and fixation.

17. Add 20 μ L of D-luciferin (15 mg/mL) per mL of SCM in the 12-well plate, gently shake the plate to mix, and incubate 5 min.

△ CRITICAL: The plate should be visually examined right after taking the plate from the incubator and before preparing for BLI acquisition. Organotypic cultures found to be not attached to their membrane or attached but with their membrane not floating are not viable and should be discarded from the analysis.

18. Acquire BLI images at exposure times of 10 and 1 s. If higher exposure times (30, 60, or 120 s) were used at day 0 BLI, then acquire additional images at the corresponding exposure times ([Figure 2C](#)).
19. Remove SCM and fix organotypic cultures by adding 1 mL of 4% paraformaldehyde (PFA) to each well ([Methods video S5](#) and [Figure 2D](#)).
 - a. Shake overnight (min. 12 h) at 4° Celsius.

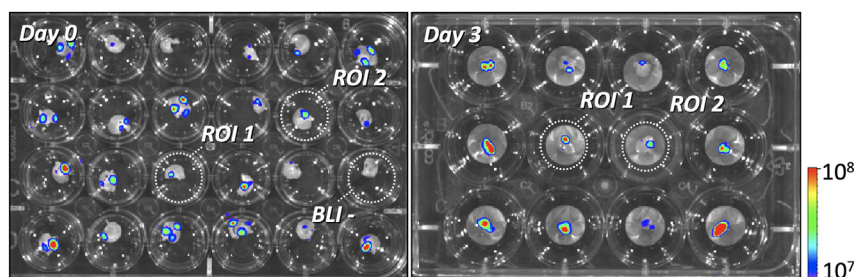


Figure 3. BLI quantification of organotypic cultures

BLI images acquired at day 0 (before plating) and day 3 (after 72 h of culture), signal is measured in photons/seconds/cm²/steradian. BLI negative (BLI-) brain slices can be found at day 0. Example of organotypic cultures with increased BLI signal or tumor growth at day 3 compared to day 0 (ROI 1) and stable BLI signal or absence of tumor growth (ROI 2) are shown.

- b. Remove 4% PFA and wash 3 times with 1 × PBS.
- c. Store fixed organotypic cultures in 1 × PBS at 4° Celsius.

Note: Steps involving 4% PFA are performed under a fume hood. For long-term storage, maintain fixed organotypic cultures in 0.25% sodium azide (in 1 × PBS) at 4° Celsius. Replace liquid periodically to avoid dry up.

⚠ **CRITICAL:** Add 4% PFA through the wall of each well and not on top of the organotypic cultures. Use the tip to push the membrane with the slice into the liquid to allow full submergence and coverage of the slice by the liquid.

EXPECTED OUTCOMES

Organotypic cultures constitute an experimental platform for brain metastasis research since they faithfully recapitulate metastatic cells *in vivo* both phenotypically and functionally,^{1,5–8} and allow rapid evaluation of multiple experimental conditions at the same time at affordable economical costs. Organotypic cultures we describe here are short-term assays that can be maintained up to 7 days with well-preserved cellular architecture and functionality of different brain resident cell types and metastatic cells able to colonize the brain.¹

This protocol describes the use of organotypic cultures of established brain metastasis to evaluate the anti-metastatic effect of chemical inhibitors through a rapid read-out of BLI. By comparing BLI signal at endpoint (i.e., day 3) and baseline (i.e., day 0), the efficacy of the chemical compound in impairing the viability of brain metastatic cells can be quantified (Figure 3), as extensively presented in our previous publication by Zhu et al.¹ Around 20–40 BLI positive brain slices can be obtained from one murine brain with established metastasis, which are sufficient to test a minimum of 4 experimental conditions plus the control per experiment. Fixed organotypic cultures can be used for complementary downstream histological analysis via immunofluorescence staining or immunohistochemistry.¹

It is important to highlight that this method allows for scalability; up to 4–5 brains can be sectioned in the vibratome at the same time, therefore, it is fully compatible with medium-throughput drug screening as we have described elsewhere.¹ In addition, we have further adapted the protocol to probe the effect of radiotherapy, another standard-of-care therapeutic modality for brain metastasis.⁵ Here, radiotherapy is applied after measurement of BLI on day 1. Importantly, administering both radiotherapy and chemical treatments, either at the same time or sequentially, is also possible.

Beyond the utility of organotypic cultures to evaluate the efficacy of various treatments on cancer cells, this method also enables researchers to interrogate the different effects of drugs on the brain

microenvironment, including toxicity and phenotypic changes. To this end the protocol can be adapted to include read-outs such as measuring lactate dehydrogenase (LDH) concentration in the supernatant of organotypic cultures or immunofluorescence staining with markers of different brain resident cell types such as reactive astrocytes and microglia, respectively.¹

Next to evaluation of treatment effects, organotypic cultures offer a variety of downstream applications that allow researchers to ask more complex questions: using unbiased omics-based approaches, the effect of therapeutics on the molecular landscape of brain metastatic cancer cells can be characterized. This requires the isolation of cancer cells from their microenvironment, by either fluorescent-activated cell sorting (FACS; e.g., based on GFP expression of cancer cells) or laser-capture microdissection, before subjecting cells to downstream processing. We show the utility of this application by interrogating the proteome of established brain metastases treated with a chaperone inhibitor, leading to the discovery of biomarkers of response as well as actionable resistance mechanisms.¹

Finally, in order to translate the versatility of the platform to human brain metastasis, the protocol described here can be adapted to generate patient-derived organotypic cultures (PDOCs) using fresh human tissue obtained from neurosurgeries.¹ This application offers researchers the ability to test treatments directly on patient material. Furthermore, we have successfully incorporated radiotherapy by applying this therapy one day after plating of PDOCs.⁵ However, some limitations are present when using patient-derived tissue. Firstly, material from surgeries is often scarce, resulting in small number of PDOCs. In addition, BLI cannot be used as an experimental read-out for cell viability as cancer cells are not engineered so they do not express luciferase. Therefore, we have adapted the protocol to use either a bromodeoxyuridine (BrdU) pulse before fixation and consequent staining of BrdU, or immunofluorescent staining of Ki67 as markers to evaluate cancer cell proliferation.¹ As a consequence, viability of the PDOCs cannot be evaluated before plating and bias in the distribution of viable PDOCs across different experimental conditions might occur. To reduce risk of bias, PDOCs from a representative cohort of patients in terms of sample size should be performed. We recommend to include PDOCs from at least 10 patients in order to get solid data.

QUANTIFICATION AND STATISTICAL ANALYSIS

BLI images of organotypic cultures are analyzed using the image analysis software Living Image 4.5 and values obtained are normalized to day 0 and the control condition.

1. Open corresponding BLI images with the software.
2. Select each well of the culture as a ROI and measure the signal in photons/seconds/cm²/steradian (Figure 3).

△ CRITICAL: Use the same ROI size to analyze all images independently of the type of culture plate used (day 0 images are obtained using 24-well plates while day 3 images are from 12-well plates). Exclude organotypic cultures that sank into the medium and were not successfully rescued.

3. For each organotypic culture, obtain the fold-increase in BLI by dividing its BLI value at day 3 by its value at day 0.

Note: Tracking of each organotypic culture during the plating phase is essential, as BLI values of the same culture at day 0 and day 3 must be used here.

4. Transform the fold-increase value of each culture into a percentage value by dividing it by the mean fold-increase value of the control condition (vehicle-treated organotypic cultures) followed by multiplication per 100.

Note: Vehicle-treated organotypic cultures with fold-increase <1 should be excluded from the calculation to obtain the mean fold-increase. When this happens, drug-treated organotypic cultures of the same region of the brain should also be excluded to avoid bias in the analysis.

5. Plot percentage values in a box and whisker plot.

Note: Mean percentage values of each experimental condition tested can be obtained to compare efficacy across treatments.

6. Test for statistical significance using the appropriate statistical test (i.e., t-test, ANOVA).

LIMITATIONS

Gentle manipulation of culture plates is critical as organotypic cultures that are found detached from the membrane or not floating when in the membrane are not viable and must be discarded from the analysis. Therefore, extreme caution should be taken when performing protocol versions requiring additional steps after plating of organotypic cultures (i.e., BLI acquisition at day 1 and application of radiation³).

Because this protocol primarily relies on BLI for read-out of treatment efficacy, heterogeneity in BLI signal across organotypic cultures from the same assay is a limiting factor. In the experimental setting, we observed that organotypic cultures with high BLI signal ($>10^7$ photons/seconds/cm²/steradian) at day 0 tend to show lower fold-increase in BLI at day 3, likely due to saturation of signal, while low BLI signal at baseline is generally associated with higher fold-increase values. Consequently, complementary histological analysis of fixed organotypic cultures¹ or additional organotypic cultures to increase sample size are often needed. Histological analyses can include immunofluorescent staining of cell proliferation markers (e.g., BrdU, Ki67) to confirm that an expected percentage of cancer cells show positive staining as per the corresponding BLI signal; organotypic cultures should be excluded otherwise. Staining of organotypic cultures from the control condition must be performed in parallel to establish reference correlations between BLI signal and percentage of proliferative cancer cells.

In the protocol we describe here, murine organotypic cultures cannot be maintained for more than 7 days without compromising tissue architecture. However, specific culture conditions that allow longer-term culturing of organotypic brain cultures do exist and have been described elsewhere,⁹ however they have been used under experimental conditions where no cancer cells were present, thus it requires optimization to be used for the purpose presented in this protocol.

Beyond technical aspects, additional limitations should be attributed to the use of organotypic cultures for brain metastasis research. They do not recapitulate steps of the metastatic disease that occur before extravasation of cancer cells from the vessels. For instance, they do not allow interrogating the ability of metastatic cells to intravasate into capillaries from the primary tumor site, to survive in circulation, and, more importantly, to cross the blood-brain barrier (BBB).

TROUBLESHOOTING

Problem 1

Embedded brain is in an undesired position after agarose is solidified (step 1f).

Potential solution

Maintain the brain in the desired position with the tip until agarose becomes slightly solid to avoid movement of the brain.

Problem 2

Embedded brain detaches from the agarose during sectioning (step 7d).

Potential solution

Gently peel away the agarose that remain attached to the brain with a forceps and embed the brain in agarose starting from step 1.

Problem 3

Plated brain slices have wrinkles (step 15).

Potential solution

Gently transfer the brain slice with the spatulas to the 10 cm plate with HBSSsup used to assist plating. Wrinkles will disappear in contact with liquid and brain slice is ready to be plated again. This procedure should be performed in the moment and not after an incubation period.

Problem 4

Membranes and/or organotypic cultures sink into the medium upon frequent movement (step 16 and subsequent movements before BLI acquisition at endpoint).

Potential solution

- Hold the edge of the membrane with a Dumont forceps and attempt to “fish” the organotypic culture to position it on top of the membrane.
- Place the membrane containing the organotypic culture between edges of the wells of the 12-well plate and leave for 30 s.
- Once the membrane looks dry, gently hold the membrane with the forceps and place it back to its corresponding well.

Note: This procedure can only rescue cultures that have just sank into the medium, approximately in the previous 30 min. For longer or unknown times, reliability is compromised, and the culture should be discarded from the analysis.

- Do not replat the organotypic culture with spatulas; this manipulation will compromise viability.

RESOURCE AVAILABILITY

Lead contact

Further information and requests for resources and reagents should be directed to and will be fulfilled by the lead contact, Manuel Valiente (mvaliente@cniio.es).

Materials availability

This protocol did not generate new unique reagents.

Data and code availability

This protocol did not generate any datasets.

SUPPLEMENTAL INFORMATION

Supplemental information can be found online at <https://doi.org/10.1016/j.xpro.2023.102194>.

ACKNOWLEDGMENTS

We want to thank I. Blanco for critical review of the manuscript. We also thank J. Massague (MSKCC) for the H2030-BrM cell line. This work was supported by MINECO (SAF2017-89643-R, SAF2014-57243-R, SAF2015-62547-ERC) (M.V.), Fundacion FERO (IX FERO Grant for Research in Oncology) (M.V.), Fundacio La Marato de TV3 (141) (M.V.), Melanoma Research Alliance (Bristol-Myers Squibb-Melanoma Research Alliance Young Investigator Award 2017 [<https://doi.org/10.48050/pc.gr.75716>]) (M.V.), Beug Foundation (Prize for Metastasis Research 2017) (M.V.), Fundacion Ramon Areces (CIVP19S8163) (M.V.), Worldwide Cancer Research (19-0177) (M.V.), H2020-FETOPEN

(828972) (M.V.), Cancer Research Institute (Clinic and Laboratory Integration Program CRI Award 2018 (54545)) (M.V.), AECC (Coordinated Translational Groups 2017 (GCTRA16015SEO) (M.V.), LAB AECC 2019 (LABAE19002-VALI) (M.V.), ERC CoG (864759) (M.V.), MICIN (PID2021-124582OB-I00) (M.V.), ASPIRE Award, The Mark Foundation (ID: 21-05-ASP), La Caixa-Severo Ochoa International PhD Program Fellowship (LCF/BQ/SO16/52270014) (L.Z.), and Boehringer Ingelheim Fonds MD fellowship (L.M.). CNIO is supported by the ISCIII, the Ministerio de Ciencia e Innovación and is a Severo Ochoa Center of Excellence (SEV-2015-0510). M.V. was named Ramon y Cajal Investigator (RYC-2013-13365) and is a member of EMBO YIP (4053).

AUTHOR CONTRIBUTIONS

M.V. and L.Z. adapted and optimized the protocol of organotypic cultures to brain metastasis. P.B. and M.P.-G. generated videos and pictures. L.Z. and L.M. wrote the manuscript. M.V. supervised the study and reviewed and edited the manuscript.

DECLARATION OF INTERESTS

The authors declare no competing interests.

REFERENCES

1. Zhu, L., Retana, D., García-Gómez, P., Álvaro-Espinosa, L., Priego, N., Masmudi-Martín, M., Yebra, N., Miarka, L., Hernández-Encinas, E., Blanco-Aparicio, C., et al. (2022). A clinically compatible drug-screening platform based on organotypic cultures identifies vulnerabilities to prevent and treat brain metastasis. *EMBO Mol. Med.* 14, e14552. <https://doi.org/10.15252/emmm.202114552>.
2. Campbell, J.P., Merkel, A.R., Masood-Campbell, S.K., Eleftheriou, F., and Sterling, J.A. (2012). Models of bone metastasis. *J. Vis. Exp.* e4260 <https://doi.org/10.3791/4260>.
3. Yardeni, T., Eckhaus, M., Morris, H.D., Huizing, M., and Hoogstraten-Miller, S. (2011). Retro-orbital injections in mice. *Lab Anim.* 40, 155–160. <https://doi.org/10.1038/labani0511-155>.
4. Nguyen, D.X., Chiang, A.C., Zhang, X.H.-F., Kim, J.Y., Kris, M.G., Ladanyi, M., Gerald, W.L., and Massagué, J. (2009). WNT/TCF signaling through LEF1 and HOXB9 mediates lung adenocarcinoma metastasis. *Cell* 138, 51–62. <https://doi.org/10.1016/j.cell.2009.04.030>.
5. Monteiro, C., Miarka, L., Perea-García, M., Priego, N., García-Gómez, P., Álvaro-Espinosa, L., de Pablos-Aragoneses, A., Yebra, N., Retana, D., Baena, P., et al. (2022). Stratification of radiosensitive brain metastases based on an actionable S100A9/RAGE resistance mechanism. *Nat. Med.* 28, 752–765. <https://doi.org/10.1038/s41591-022-01749-8>.
6. Er, E.E., Valiente, M., Ganesh, K., Zou, Y., Agrawal, S., Hu, J., Griscom, B., Rosenblum, M., Boire, A., Brogi, E., et al. (2018). Pericyte-like spreading by disseminated cancer cells activates YAP and MRTF for metastatic colonization. *Nat. Cell Biol.* 20, 966–978. <https://doi.org/10.1038/s41556-018-0138-8>.
7. Valiente, M., Obenauf, A.C., Jin, X., Chen, Q., Zhang, X.H.-F., Lee, D.J., Chaff, J.E., Kris, M.G., Huse, J.T., Brogi, E., and Massagué, J. (2014). Serpins promote cancer cell survival and vascular co-option in brain metastasis. *Cell* 156, 1002–1016. <https://doi.org/10.1016/j.cell.2014.01.040>.
8. Priego, N., Zhu, L., Monteiro, C., Mulders, M., Wasilewski, D., Bindeman, W., Doglio, L., Martínez, L., Martínez-Saez, E., Ramón Y Cajal, S., et al. (2018). STAT3 labels a subpopulation of reactive astrocytes required for brain metastasis. *Nat. Med.* 24, 1024–1035. <https://doi.org/10.1038/s41591-018-0044-4>.
9. De Simoni, A., and Yu, L.M.Y. (2006). Preparation of organotypic hippocampal slice cultures: interface method. *Nat. Protoc.* 1, 1439–1445. <https://doi.org/10.1038/nprot.2006.228>.